

evaporative cooling. We also plotted photon scattering rates under a constant potential energy of $100 \mu\text{K}$ for atoms in $|F = 6, m_F = -6\rangle$ lowest Zeeman state with π polarization light¹. One can see that laser light at a longer wavelength is more preferable for evaporative cooling.

In this experiment, We used a 1550 nm laser light for ODT beams. The polarizability and photon scattering rate are listed in Table 6.1.

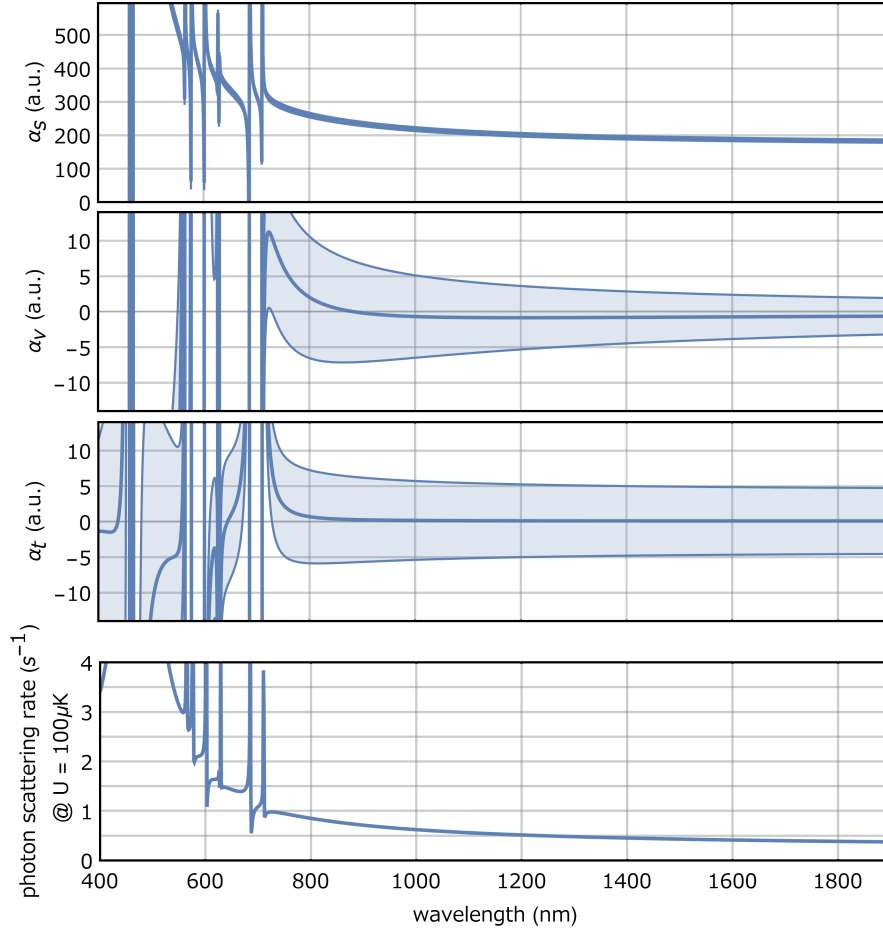


Figure 6.2: Scalar, vector, and tensor polarizability in the ground state as a function of the optical dipole trap wavelength. The shade region represents the uncertainty of the polarizabilities, which comes from uncertainties of the transition probabilities used in this calculation. The bottom figure shows photon scattering rate under a constant potential energy of $100 \mu\text{K}$ for atoms in $|F = 6, m_F = -6\rangle$ lowest Zeeman state with π polarization light.

¹Although the photon scattering rate depends on Zeeman sublevel and polarization of light, their contribution is quite small since imaginary part of vector and tensor polarizability is also much smaller than scalar polarizability (within a few percentage of the scalar polarizability) except for near resonant wavelengths. It is in contrast to Er and Dy atoms [69, 72], which have large electric orbital angular momentum in their ground state.

Wavelength	α_s	α_v	α_t	photon scattering rate @ $U = 100 \mu\text{K}$
1550 nm	189(5) a.u.	0(3) a.u.	0(5) a.u.	0.43 s^{-1}

Table 6.1: Scalar, vector and tensor polarizability for 1550 nm light. The polarizability is given in atomic units of $4\pi\epsilon_0 a_0^3$, where a_0 is the Bohr radius. The photon scattering rate under a potential energy of $100 \mu\text{K}$ is also listed.

6.2 Setup for an ODT

In ODT experiment for europium, we use 1550 nm laser with a maximum power of 15 W (see Sec. 4.2.7). We adopt crossed ODT geometry [73], whose setup is illustrated in Fig. 6.3. The trap consists of two crossing beams, a tightly confining horizontal beam and a less focused vertical beam. The beam parameters are summarized in Table 6.2.

The basic idea for evaporative cooling in the crossed ODT geometry is as follows. We first load atoms from red-MOT into horizontal ODT. In this condition atomic density is not sufficiently high to conduct evaporative cooling since confinement of the horizontal ODT along axial direction is quite weak. Then, we add vertical ODT and gradually reduce power of horizontal ODT. Although almost atoms are initially trapped in the horizontal ODT and spread along axial direction, the atoms become concentrated into the crossed region as the horizontal ODT power decreases. Through this evaporative cooling procedure, we increase phase space density of an Eu atomic cloud and obtain a BEC.

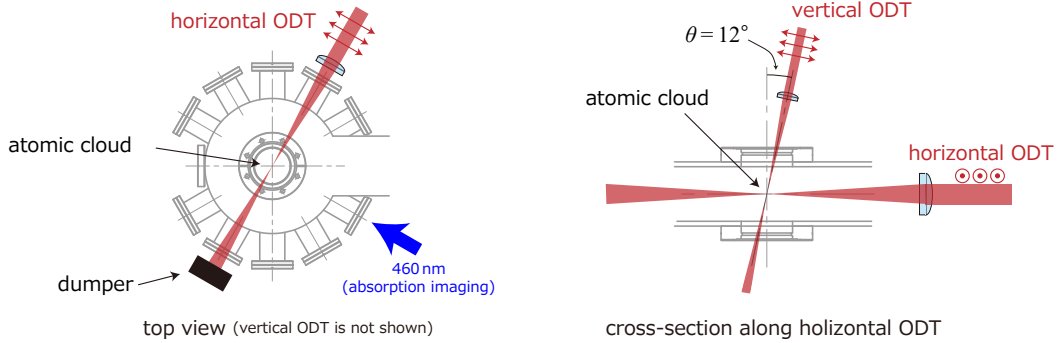


Figure 6.3: Setup for ODT.

6.3 Loading to an ODT

As described in previous section, we loaded atoms from the red-MOT to the horizontal ODT. After loading atoms to the red-MOT, we turned off laser lights that are not necessary for retaining red-MOT and blocked an atomic beam. Then we compressed red-MOT with following parameters: $\partial B_z / \partial z = 2 \times 10^{-2} \text{ T/m}$, $I_{687} = 0.8 I_{s,687}$, and $\delta_{687} = -21 \Gamma_{687}$. The compressed red-MOT contains 3×10^7 atoms with a temperature of $\sim 6 \mu\text{K}$. Subsequently, we turned on the horizontal ODT beam with a power of 11.5 W, which overlaps the com-

	beam waist	maximum power	maximum potential depth
horizontal ODT	26 (30) μm	11.5 W	400 μK
vertical ODT	47 μm	7.4 W	90 μK

Table 6.2: List of ODT parameters. The beam waists are measured by Thorlabs’ scanning-slit optical beam profiler BP209-IR/M. The beam profile of horizontal ODT has an elliptic cross section, where minor (major) axis is vertical (horizontal) direction. Note that the maximum power is available only when full power of the laser is assigned to one beam (see power recycling system in Sec. 4.2.7). Maximum potential depth is calculated based on the maximum power, the beam waist, and polarizability listed in Table 6.1.

pressed red-MOT. After 40 ms of loading, we obtained 1.4×10^6 atoms with a temperature of 50 μK in the horizontal ODT, whose absorption image is shown in Fig 6.4.

We note that the loading efficiency strongly depends on the polarization of the horizontal ODT beam. The successful loading was achieved with the horizontally polarized ODT beam, whereas we loaded only 2×10^5 atoms in the case of a vertically polarized ODT beam. The reason could be explained by differential tensor light shift for the 687 nm cooling transition, which is investigated in Sec. 6.4.

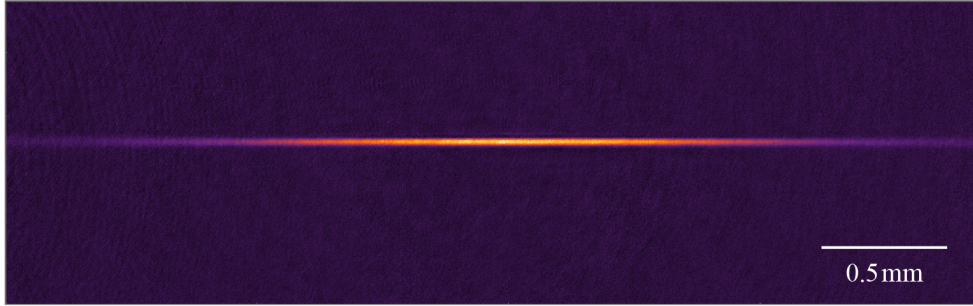


Figure 6.4: Absorption image after loading atoms to horizontal ODT

The loading efficiency from red-MOT to the ODT is only $\sim 5\%$ at best. The low loading efficiency might be due to size mismatch between the horizontal ODT beam ($\sim 30 \mu\text{m}$ beam waist) and bowl-shaped compressed red-MOT ($\sim 2 \text{ mm}$ diameter). Although one can increase the overlapping volume by expanding the ODT beam, which results in a reduction of the trap confinement force and less efficient evaporative cooling. One of the solution is a tunable trapping volume system using an AOM, which was introduced for other dipolar gas experiments [74–76]. Increasing loading efficiency will be a future task.

6.4 Differential light shift measurement

6.4.1 Differential light shift

Differential light shift is the difference between the light shifts for the excited state $U_e(r)$ and the ground state $U_g(r)$ in Eq. 6.1, which is represented as

$$\Delta U(r) = U_e(r) - U_g(r)$$

6.5

7.1 Creation of a Eu BEC

7.1.1 Evaporative cooling

To obtain a BEC, we have to increase phase space density (PSD) of trapped atoms, which is defined as

$$\text{PSD} = n_0 \lambda_{dB}^3 \quad 7.1$$

where n_0 is the peak number density of the gas, $\lambda_{dB} = h/\sqrt{2\pi m k_B T}$ is the thermal de Broglie wavelength, m is the atomic mass of Eu, k_B is the Boltzmann constant, and T is the temperature of the trapped atoms. Both high atom number density and low temperature are required for high PSD. The phase transition to BEC occurs when the PSD reaches to 2.6 [78], where the thermal de Broglie wavelength is on the order of the interparticle distance.

As described in Sec. 6.3, initial condition just after loading to horizontal ODT is 1.4×10^6 atoms at a temperature of $50 \mu\text{K}$, which corresponds to a PSD of only 5×10^{-6} . We increased the PSD with evaporative cooling technique. We gradually decreased the trap depth and let the most energetic atoms in the trap escape over the potential barrier. The evaporative cooling was achieved in the following sequence. After loading atoms to the horizontal ODT, we tuned off the red-MOT beams and the quadrupole magnetic field. Then, we switched on a homogeneous magnetic field of 1.1 G along the horizontal ODT beam to preserve the spin polarization of the sample. Subsequently, we gradually decreased horizontal ODT power and increase the vertical ODT power, as described in Fig. 7.1. This operation concentrated atoms into the crossed region and increased atom number density; almost all atoms were trapped in the crossed region at the evaporation time of 6 s. After 7.4 s evaporative cooling we produced an almost pure BEC with 3×10^3 atoms, which is described in the next section.

Note that the BEC was obtained only when the polarization of the vODT was orthogonal to that of the hODT. When the polarization of the vODT was parallel, atoms seems to be heated even though the frequencies of the two trapping beam differs 160 MHz. Further investigation is required to clarify the cause of the polarization dependence.

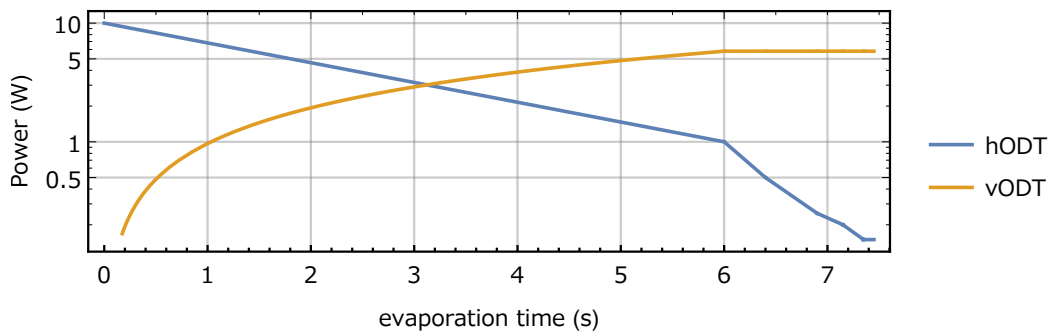


Figure 7.1: Crossed ODT beam power trajectory during evaporative cooling.

7.1.2 First observation of a Eu BEC

Figure 7.2 shows the absorption images taken after 13 ms of expansion for different final temperature. At higher temperature (a) the atomic distribution is well described by Maxwell-Boltzmann distribution. By cooling the atoms below the critical temperature, we observed a clear bimodal profile (b). The characteristic shape was one of the key evidence for BEC. At lower temperature, we observed almost pure condensate with 3×10^3 atoms (c). The slightly tilting elliptic shape of the condensate reflects the vODT beam crossed to hODT with an angle of 78° , which is illustrated in the absorption image (c).

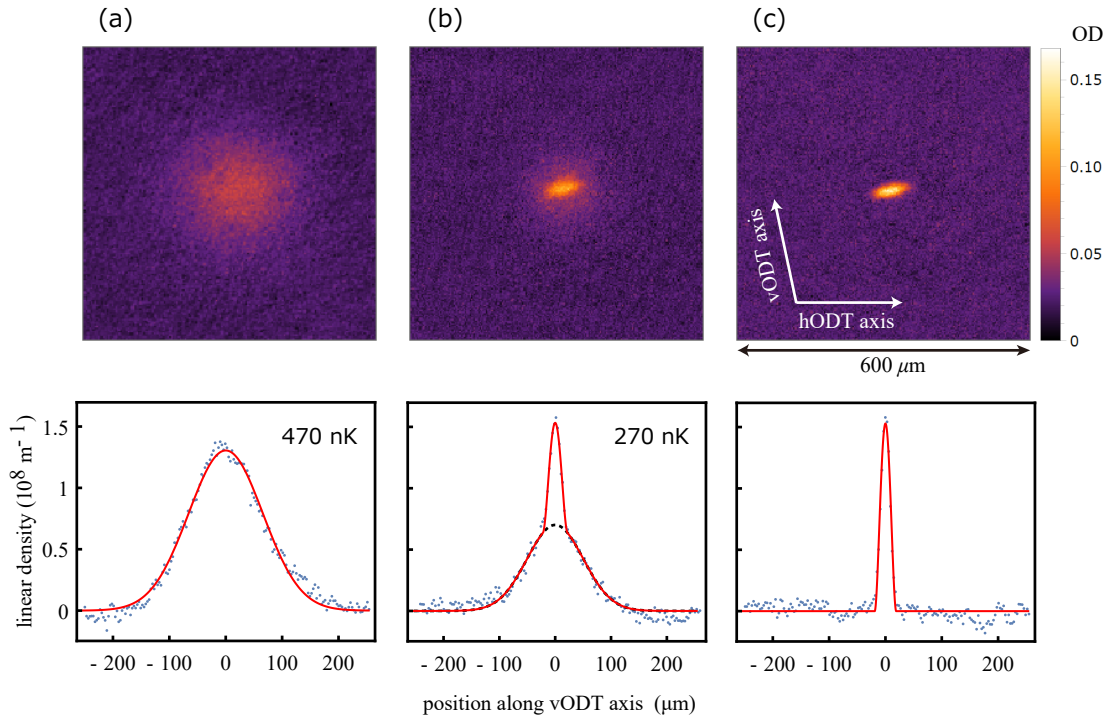


Figure 7.2: (Top) Absorption images showing the BEC phase transition for different final hODT power. The absorption images are an average of five images taken after 13 ms of expansion. (Bottom) Integrated linear profiles along direction orthogonal to vODT axis. The solid lines are fit to the data using Gaussian (a), bimodal (b), and Thomas-Fermi (c) distribution. The dotted lines represent the Gaussian part of the bimodal fit. From the fit we extract: $N = 2.2 \times 10^4$, $T = 470$ nK (a), $N = 1.1 \times 10^4$, $T = 270$ nK (b), and $N = 3 \times 10^3$ (c), where N is the total atom number.

To confirm the validity of the phase transition, we calculated a critical temperature. The critical temperature is given by the following equation: [79]

$$T_C^0 = \frac{\hbar\bar{\omega}}{k_B} \left(\frac{N}{\xi(3)} \right)^{\frac{1}{3}} \simeq 0.94 \frac{\hbar\bar{\omega}}{k_B} N^{\frac{1}{3}} \quad 7.2$$

Here, $\bar{\omega}$ is the geometric mean trap frequency, which we extracted from ODT beam waists and power, and polarizability calculated in Sec. 6.1. When finite size effects as well as a correction arising from the contact interaction [79] are taken into account, the critical

temperature is shifted to lower temperatures:

$$T_C^* = T_C^0 + \delta T_C^{fs} + \delta T_C^{int} \quad 7.3$$

where $\delta T_C^{fs} \simeq -0.73 \frac{\langle \omega \rangle}{\bar{\omega}} N^{-\frac{1}{3}} T_C^0$ is a shift due to the finite number of atoms and $\delta T_C^{int} \simeq -1.33 \frac{a_s}{a_{HO}} N^{\frac{1}{6}} T_C^0$ is a shift due to contact interaction. Here $\langle \omega \rangle$ is the arithmetic mean trap frequency, a_s is the s-wave scattering length, $a_{HO} = \sqrt{\frac{\hbar}{m\bar{\omega}}}$ is the length of the harmonic oscillator corresponding to the trap. We set $a_s = 135 a_B$, that is measured in Sec. 7.2.3. By using aforementioned equations, we obtain $T/T_C^* = 1.14$ (a) and $T/T_C^* = 0.86$ (b), which are consistent with our observation.

7.1.3 Optimization of the evaporative cooling

After observation of the BEC, we optimized the evaporative cooling process, increasing a number of condensed atoms to 1.5×10^4 . The key ingredients for the improvement were loosening vODT confinement in the last stage of the evaporative cooling, and setting the bias magnetic field to 3.0 G during the evaporative cooling. Figure 7.3 shows improved crossed ODT beam power trajectory. The power of vODT was reduced to 1/4 in the last stage of the evaporative cooling, which reduces three-body inelastic collision.

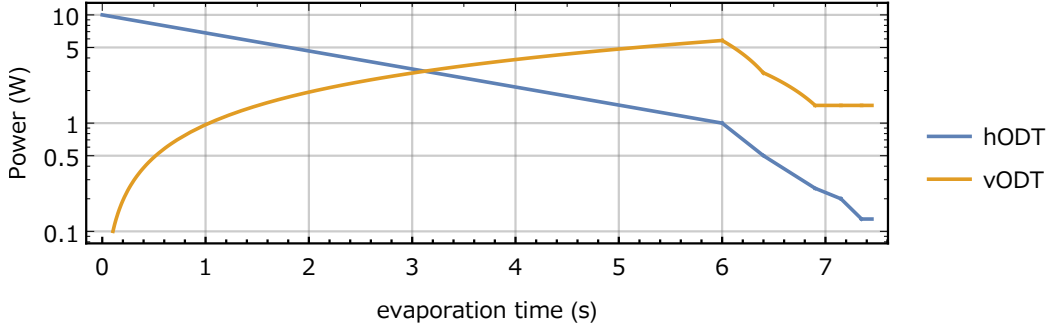


Figure 7.3: Improved crossed ODT beam power trajectory during evaporative cooling.

A value of the bias magnetic field is also important for efficient evaporative cooling in the case of Eu. Figure 7.4 shows number of atoms after evaporative cooling as a function of bias magnetic field during the evaporative cooling. Here the magnetic field was oriented to vertical direction and almost all atoms were condensed at the magnetic field where larger number of atoms were remained. One can see clear dependence of the remained atoms on the bias magnetic field. It would be due to inelastic collisional loss induced by Feshbach resonances. Maximum number of atoms was obtained around a magnetic field of 3.0 G.

The evaporative cooling efficiency is often evaluated by using the following formula [80]:

$$\gamma = -\frac{\ln(\rho_f/\rho_i)}{\ln(N_f/N_i)} \quad 7.4$$

where ρ_i , ρ_f are the initial and final PSD, and N_i , N_f are the initial and final number of atoms. The γ represents the order of magnitude increase of the PSD at the expense of an order of magnitude of atom number. By using this formula, overall efficiency of our

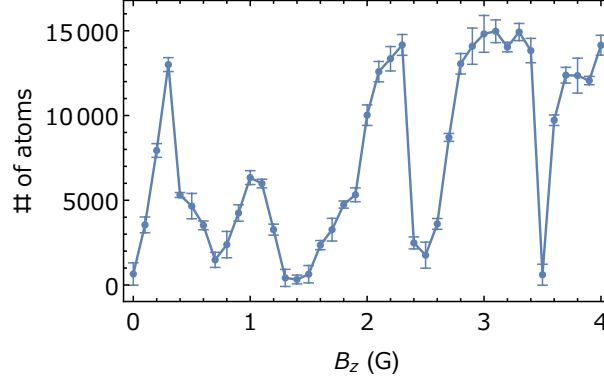


Figure 7.4: Number of atoms after evaporative cooling as a function of bias magnetic field.

evaporative cooling process was evaluated to be $\gamma = 3.6$, which is a good value considering many other experiments typically achieve values of $\gamma \approx 2.5 \sim 3.5$ [80].

7.1.4 Lifetime of the BEC

The lifetime of the BEC in the trap is of interest since one can manipulate or develop the BEC within the time. We measured the number of atoms as a function of the holding time after creating the BEC, which is shown in Fig. 7.5. The data was fitted using a three-body and one-body loss model for a three dimensional BEC [81]:

$$\frac{dN}{dt} = -\gamma N^{9/5} - \alpha N \quad 7.5$$

whose solution is given by:

$$N(t) = \left[\left(N_0^{-4/5} + \frac{\gamma}{\alpha} \right) e^{\frac{4}{5}\alpha t} - \frac{\gamma}{\alpha} \right]^{-5/4} \quad 7.6$$

The data was described by only three-body loss, and we couldn't estimate one-body loss rate α . A $1/e$ lifetime for the initial three-body loss is estimated to $(\gamma N_0^{4/5})^{-1} = 1.4$ s. Corresponding three-body loss coefficient can be obtained by the following [81]:

$$L = \gamma \frac{6}{7} \left((15^{2/5}/(14\pi)) (m\bar{\omega}/(\hbar\sqrt{a_s}))^{6/5} \right)^{-2} \quad 7.7$$

By using $a_s = 135 a_B$ and $\bar{\omega} = 2\pi \times 204$ Hz, we extract $L \sim 10^{-29}$ cm⁶/s, which is the same order of that for rubidium atom in $|F = 2, m_F = 2\rangle$ [81].

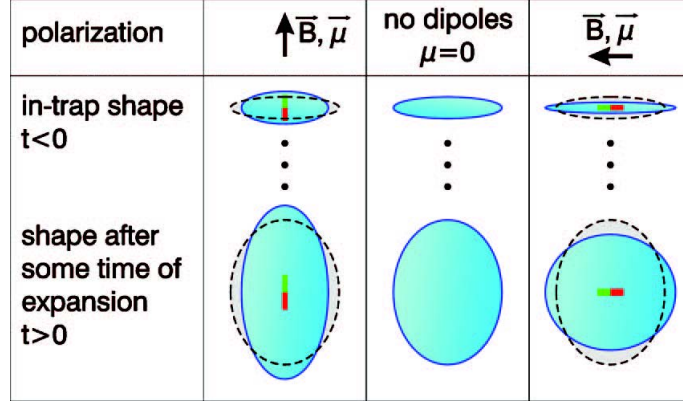


Figure 7.6: Figure to illustrate the change of the condensate shape under the influence of magnetization in trap (top row) and during time of flight (bottom row). Left column, magnetization in transversal direction; center column, nondipolar atoms; right column, longitudinal polarization. Dashed ellipses represent the nondipolar condensate. Reprinted from Ref. [83].

7.2.2 Observation of dipole-dipole interaction in a BEC

As described in the previous section, DDI effect can be observed as an anisotropic deformation of the expanding BEC. To confirm the DDI effect of the Eu BEC, we took two absorption images of BECs that were released under a bias magnetic field of 1.5 G oriented to vertical and horizontal direction. The images are shown in Fig. 7.7. One can see that the shape of the expanding BEC is clearly elongated to the direction of the magnetic field. It is a clear evidence that Eu BEC is a strongly dipolar BEC. The ratio between the width of the minor axis in (a) and (b) reaches to 1.25.

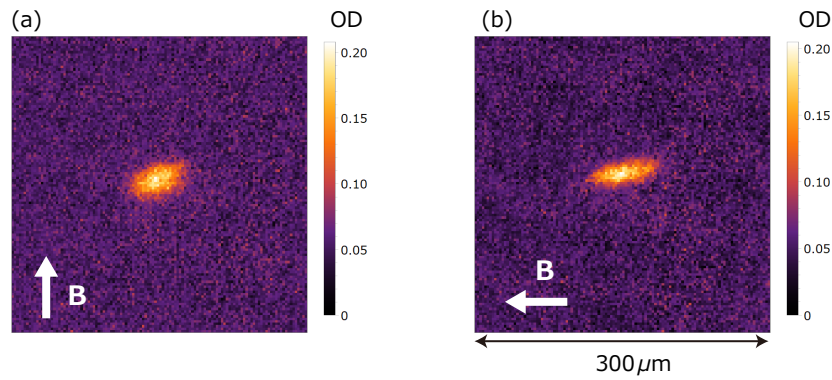


Figure 7.7: Absorption images of a BEC taken after 13 ms of expansion. The image shows the BEC that was released under a bias magnetic field of 1.5 G oriented to vertical (a) and horizontal (b) direction. White arrows indicate the magnetic field direction.

7.2.3 Estimation of s-wave scattering length

The s-wave scattering length is a crucial parameter that characterizes the BEC. A simple method to measure the s-wave scattering length is to estimate the scattering length from an expansion speed of the BEC [73, 89]. This method is also applicable for dipolar

BEC [4]. However, the estimation often accompanies large errors due to error of number of atoms or expansion speed.

Axel Griesmaier proposed and demonstrated a simple and more accurate method to estimate the s-wave scattering length in dipolar BEC [4]. He measured asymptotic velocities of expansion of the ^{52}Cr condensate with different orientation of the atomic magnetic moment. Then, he compared the ratio of the velocities to the numerical simulation of the expanding dipolar BEC and got a relative DDI strength ϵ_{dd} and corresponding s-wave scattering length. The values was in excellent agreement with the s-wave scattering length that has been obtained by comparing the measured positions of Feshbach resonance in chromium collisions with multichannel calculations [90].

Here we use the same method to estimate s-wave scattering length. In the previous section, we already got a experimental result on the ratio of the velocities of 1.25. By comparing the ratio to the numerical solution, we get a $\epsilon_{dd} = 0.44$, which corresponds to a s-wave scattering length of $a_s = 135 a_B$. Note that the magnetic field direction is slightly tilted to the ODT trapping axis ($\sim 12^\circ$), which I did not take into account to the calculation. More proper estimation is a future task.

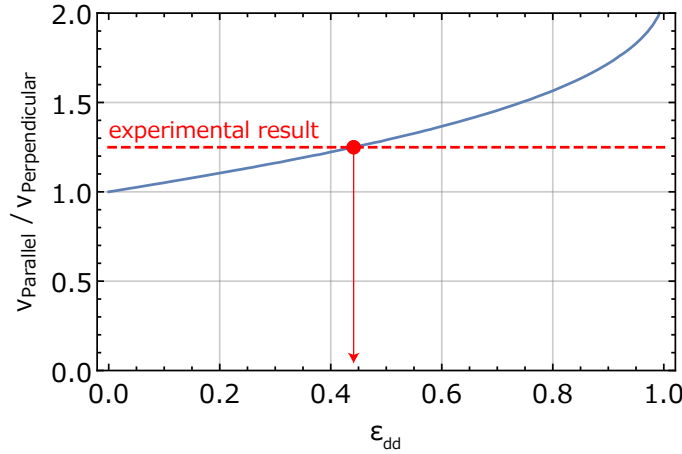


Figure 7.8: Estimation of the s-wave scattering length for ^{151}Eu . The solid blue line is a numerical simulation of the ratio of the expanding velocities when the magnetic field was set to parallel and perpendicular to the velocity axis, as a function of ϵ_{dd} . Dashed red line indicates experimental result. From the crossing point of the experimental result and numerical simulation, we extracted a relative DDI strength of $\epsilon_{dd} = 0.44$.

7.3 Feshbach resonance

Feshbach resonance is a powerful tool to control the interaction between ultracold atoms [91]. The experiment on Feshbach resonance is currently in progress. In this section, I will present some preliminary results of the Feshbach spectroscopy for Eu atoms.

7.3.1 Feshbach spectroscopy ($0 \sim 4 \text{ G}$)

In the optimization of the evaporative cooling in Sec. 7.1.3, we observed that evaporative cooling efficiency depends on a value of bias magnetic field, which suggests there are